

Unit 2. Vector Spaces, Basis, Dimension

1 Vector spaces and subspaces

1.1 Definition and examples

Definition 2.1 (Real vector space). A **real vector space** is a nonempty set $\mathcal{V}$ with operations

$$+ : \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}, \quad \cdot : \mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V},$$

satisfying, for $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{R}$:

- | | |
|--|---|
| (i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$; | (v) $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$; |
| (ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$; | (vi) $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$; |
| (iii) there is $\mathbf{0} \in \mathcal{V}$ with $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every $\mathbf{v}$; | (vii) $(\alpha\beta)\mathbf{v} = \alpha(\beta\mathbf{v})$; |
| (iv) every $\mathbf{v}$ has $-\mathbf{v} \in \mathcal{V}$ with $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$; | (viii) $1\mathbf{v} = \mathbf{v}$. |

Members of $\mathcal{V}$ are **vectors**; elements of $\mathbb{R}$ are **scalars**.

Origin: S §1.1.1

Theorem 2.2 (Consequences of the vector-space axioms). In a real vector space:

- (i) the zero vector and the additive inverse of each vector are unique;

- (ii) $0\mathbf{v} = \mathbf{0}$ and $\alpha\mathbf{0} = \mathbf{0}$;
- (iii) $(-1)\mathbf{v} = -\mathbf{v}$ and $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v}) = \alpha(-\mathbf{v})$;
- (iv) $\mathbf{u} + \mathbf{w} = \mathbf{v} + \mathbf{w}$ implies $\mathbf{u} = \mathbf{v}$.
- (v) $\alpha\mathbf{v} = \mathbf{0}$ if and only if $\alpha = 0$ or $\mathbf{v} = \mathbf{0}$.

Origin: A&M Exs. 3.2–3.3

Proof.

$$\mathbf{0}_1 = \mathbf{0}_1 + \mathbf{0}_2 = \mathbf{0}_2,$$

and, if $\mathbf{w}_1 + \mathbf{v} = \mathbf{w}_2 + \mathbf{v} = \mathbf{0}$,

$$\mathbf{w}_1 = \mathbf{w}_1 + \mathbf{0} = \mathbf{w}_1 + (\mathbf{v} + \mathbf{w}_2) = (\mathbf{w}_1 + \mathbf{v}) + \mathbf{w}_2 = \mathbf{0} + \mathbf{w}_2 = \mathbf{w}_2.$$

$$0\mathbf{v} = (0 + 0)\mathbf{v} = 0\mathbf{v} + 0\mathbf{v} \implies 0\mathbf{v} = \mathbf{0},$$

$$\alpha\mathbf{0} = \alpha(\mathbf{0} + \mathbf{0}) = \alpha\mathbf{0} + \alpha\mathbf{0} \implies \alpha\mathbf{0} = \mathbf{0}.$$

Each implication adds the appropriate additive inverse. Moreover,

$$\begin{aligned}(-1)\mathbf{v} + \mathbf{v} &= (-1 + 1)\mathbf{v} = \mathbf{0}, \\ (-\alpha)\mathbf{v} + \alpha\mathbf{v} &= (-\alpha + \alpha)\mathbf{v} = \mathbf{0}, \\ \alpha(-\mathbf{v}) &= \alpha((-1)\mathbf{v}) = (-\alpha)\mathbf{v}.\end{aligned}$$

Uniqueness of additive inverses gives (iii), and

$$\mathbf{u} + \mathbf{w} = \mathbf{v} + \mathbf{w} \implies (\mathbf{u} + \mathbf{w}) - \mathbf{w} = (\mathbf{v} + \mathbf{w}) - \mathbf{w} \implies \mathbf{u} = \mathbf{v}.$$

If $\alpha\mathbf{v} = \mathbf{0}$ and $\alpha \neq 0$, then

$$\mathbf{v} = \alpha^{-1}(\alpha\mathbf{v}) = \alpha^{-1}\mathbf{0} = \mathbf{0}.$$

Conversely,

$$\alpha = 0 \implies \alpha\mathbf{v} = 0\mathbf{v} = \mathbf{0}, \quad \mathbf{v} = \mathbf{0} \implies \alpha\mathbf{v} = \alpha\mathbf{0} = \mathbf{0}.$$



Example 2.1 (Coordinate spaces). With componentwise operations, $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ are vector spaces; their zero vectors are $\mathbf{0}$ and $\mathbf{O}$. Each axiom is the corresponding scalar identity applied componentwise.

Origin: A&M Ex. 3.4; H §4.1b

Example 2.2 (Polynomials and functions). For a nonnegative integer d , let

$$\mathcal{P}_d = \{a_0 + a_1t + \cdots + a_dt^d : a_0, \dots, a_d \in \mathbb{R}\}.$$

Equality in $\mathcal{P}_d$ means coefficientwise equality; addition and scalar multiplication are coefficientwise. More generally, if D is any nonempty set, the real-valued functions on D form a vector space under

$$(f + g)(x) = f(x) + g(x), \quad (\alpha f)(x) = \alpha f(x).$$

Equality and every axiom are checked at each $x \in D$. The space $\mathcal{P}_d$ is closed under these operations because adding or scaling two polynomials of degree at most d cannot introduce a higher power of t .

Origin: Course formulation from S §1.1.1

1.2 Subspaces

Definition 2.3 (Subspace). Let $\mathcal{V}$ be a vector space. A nonempty subset $\mathcal{S} \subseteq \mathcal{V}$ is a **subspace** of $\mathcal{V}$ if

$$\mathbf{u}, \mathbf{v} \in \mathcal{S} \implies \mathbf{u} + \mathbf{v} \in \mathcal{S}, \quad \alpha \in \mathbb{R}, \mathbf{u} \in \mathcal{S} \implies \alpha \mathbf{u} \in \mathcal{S}.$$

It uses the addition and scalar multiplication already defined on $\mathcal{V}$.

Origin: H §4.2; A&M Ex. 3.8

Theorem 2.4 (Subspace test). A nonempty subset $\mathcal{S}$ of a real vector space $\mathcal{V}$ is a subspace if and only if

$$\alpha \mathbf{u} + \beta \mathbf{v} \in \mathcal{S} \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathcal{S} \text{ and } \alpha, \beta \in \mathbb{R}.$$

Origin: H §4.1b; A&M Ex. 3.8

Proof. For $\mathbf{u}, \mathbf{v} \in \mathcal{S}$,

$$\mathcal{S} \text{ subspace} \implies \alpha \mathbf{u}, \beta \mathbf{v} \in \mathcal{S} \implies \alpha \mathbf{u} + \beta \mathbf{v} \in \mathcal{S}.$$

Conversely, the asserted closure with $(\alpha, \beta) = (1, 1)$ and $(a, 0)$ gives

$$\mathbf{u} + \mathbf{v} \in \mathcal{S}, \quad a\mathbf{u} + 0\mathbf{v} = a\mathbf{u} \in \mathcal{S}.$$

□

Example 2.3 (Homogeneous and affine constraints). In $\mathcal{P}_2$, let

$$\mathcal{S} = \{p : p(1) = 0\}, \quad \mathcal{A} = \{p : p(1) = 1\}.$$

The zero polynomial belongs to $\mathcal{S}$, and if $p, q \in \mathcal{S}$, then

$$(\alpha p + \beta q)(1) = \alpha p(1) + \beta q(1) = 0.$$

Thus $\mathcal{S}$ is a subspace, whereas $0 \notin \mathcal{A}$.

Origin: A&M Exs. 3.9–3.10

Example 2.4 (Symmetric matrices). The symmetric matrices form a subspace of $\mathbb{R}^{n \times n}$. The null

matrix is both symmetric and skew-symmetric, and for symmetric $\mathbf{A}, \mathbf{B}$,

$$(\alpha\mathbf{A} + \beta\mathbf{B})' = \alpha\mathbf{A}' + \beta\mathbf{B}' = \alpha\mathbf{A} + \beta\mathbf{B}.$$

For skew-symmetric $\mathbf{A}, \mathbf{B}$, the same calculation gives $(\alpha\mathbf{A} + \beta\mathbf{B})' = -(\alpha\mathbf{A} + \beta\mathbf{B})$, so the skew-symmetric matrices form another subspace.

Origin: H §4.3b; A&M Ex. 3.11

Theorem 2.5 (Intersections and unions). The intersection of any nonempty collection of subspaces of $\mathcal{V}$ is a subspace. For two subspaces $\mathcal{S}$ and $\mathcal{T}$, the union $\mathcal{S} \cup \mathcal{T}$ is a subspace if and only if $\mathcal{S} \subseteq \mathcal{T}$ or $\mathcal{T} \subseteq \mathcal{S}$.

Origin: S §1.1.1; A&M Ex. 3.12(a); H Ex. 4.3

Proof. For a nonempty family $\{\mathcal{S}_i : i \in I\}$,

$$\mathbf{0} \in \bigcap_i \mathcal{S}_i, \quad \mathbf{u}, \mathbf{v} \in \bigcap_i \mathcal{S}_i \implies (\forall i) \alpha\mathbf{u} + \beta\mathbf{v} \in \mathcal{S}_i \implies \alpha\mathbf{u} + \beta\mathbf{v} \in \bigcap_i \mathcal{S}_i.$$

If one of $\mathcal{S}, \mathcal{T}$ contains the other, their union is the larger subspace. For the converse, take $\mathbf{s} \in \mathcal{S} \setminus \mathcal{T}$

and $\mathbf{t} \in \mathcal{T} \setminus \mathcal{S}$. Closure of the union would give $\mathbf{s} + \mathbf{t} \in \mathcal{S} \cup \mathcal{T}$, whereas

$$\mathbf{s} + \mathbf{t} \in \mathcal{S} \implies \mathbf{t} = (\mathbf{s} + \mathbf{t}) - \mathbf{s} \in \mathcal{S}, \quad \mathbf{s} + \mathbf{t} \in \mathcal{T} \implies \mathbf{s} = (\mathbf{s} + \mathbf{t}) - \mathbf{t} \in \mathcal{T}.$$



1.3 Span and linear independence

Definition 2.6 (Span). For a finite list $\mathbf{v}_1, \dots, \mathbf{v}_k$ in $\mathcal{V}$, its **span** is

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \left\{ \sum_{j=1}^k \alpha_j \mathbf{v}_j : \alpha_1, \dots, \alpha_k \in \mathbb{R} \right\}.$$

The empty sum is $\mathbf{0}$, so $\text{span } \emptyset = \{\mathbf{0}\}$.

Origin: H §4.3a

Theorem 2.7 (The span is the smallest containing subspace). The set $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is a subspace of $\mathcal{V}$. If a subspace $\mathcal{S}$ contains every $\mathbf{v}_j$, then it contains their span.

Origin: H Ex. 4.5; A&M Ex. 3.18

Proof. Taking every coefficient zero gives $\mathbf{0}$ in the span. If $\mathbf{u} = \sum_j a_j \mathbf{v}_j$ and $\mathbf{w} = \sum_j b_j \mathbf{v}_j$, then

$$\alpha \mathbf{u} + \beta \mathbf{w} = \sum_{j=1}^k (\alpha a_j + \beta b_j) \mathbf{v}_j,$$

so the span is a subspace. If $\mathbf{v}_j \in \mathcal{S}$ for every j , repeated closure gives

$$(\alpha_1, \dots, \alpha_k) \in \mathbb{R}^k \implies \sum_{j=1}^k \alpha_j \mathbf{v}_j \in \mathcal{S}.$$

□

Definition 2.8 (Linear independence). A finite list $\mathbf{v}_1, \dots, \mathbf{v}_k$ is **linearly independent** if

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0} \implies \alpha_1 = \dots = \alpha_k = 0.$$

It is **linearly dependent** otherwise. The empty list is independent.

Theorem 2.9 (Dependence and span). A nonempty finite list is linearly dependent if and only if one of its members belongs to the span of the others.

Origin: H Lemma 3.2.1

Proof. For coefficients not all zero, choose j with $\alpha_j \neq 0$:

$$\sum_i \alpha_i \mathbf{v}_i = \mathbf{0} \implies \mathbf{v}_j = - \sum_{i \neq j} \frac{\alpha_i}{\alpha_j} \mathbf{v}_i.$$

Conversely,

$$\mathbf{v}_j = \sum_{i \neq j} c_i \mathbf{v}_i \implies \mathbf{v}_j - \sum_{i \neq j} c_i \mathbf{v}_i = \mathbf{0},$$

a nontrivial zero combination. □

Corollary 2.10 (Appending one vector). For any finite list $\mathbf{v}_1, \dots, \mathbf{v}_k$,

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$$

if and only if $\mathbf{w}$ belongs to the span on the right. If the original list is independent, then the

appended list is independent if and only if $\mathbf{w}$ does not belong to that span.

Origin: H Cor. 3.2.3 and Lemma 4.3.1

Proof. Put $\mathcal{S} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. Since $\mathcal{S} \subseteq \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}\}$,

$$\mathbf{w} \in \mathcal{S} \iff \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}\} \subseteq \mathcal{S} \iff \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}\} = \mathcal{S}.$$

If the original list is independent, then

$$\sum_{j=1}^k a_j \mathbf{v}_j + c\mathbf{w} = \mathbf{0}, \quad (a_1, \dots, a_k, c) \neq \mathbf{0} \iff c \neq 0, \quad \mathbf{w} = -\sum_{j=1}^k (a_j/c) \mathbf{v}_j.$$

□

Example 2.5 (A redundant list). In $\mathbb{R}^3$, let

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

The first two vectors are independent: the first two components of $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{0}$ give $a = b = 0$. Since $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$, the three-vector list is dependent and has the same span as $\mathbf{v}_1, \mathbf{v}_2$. That span is the plane

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : z = x + y \right\}.$$

Origin: Course formulation from H Lemmas 3.2.1 and 4.3.1

2 Basis and dimension

2.1 Basis

Definition 2.11 (Basis and finite-dimensional space). A finite list in $\mathcal{V}$ is a **basis** of $\mathcal{V}$ if it is linearly independent and spans $\mathcal{V}$. The space $\mathcal{V}$ is **finite-dimensional** if some finite list spans it. The empty list is a basis of the zero space $\{\mathbf{0}\}$.

Origin: H §§4.3a and 4.3e

Theorem 2.12 (Extracting and extending a basis). Let $\mathcal{V}$ be finite-dimensional.

- (i) Every finite spanning list contains a sublist that is a basis of $\mathcal{V}$.
- (ii) Every finite independent list can be extended to a basis of $\mathcal{V}$.

Origin: H Thms. 4.3.11–4.3.12

Proof. For (i), repeat

dependent spanning list $\mapsto$ list with one redundant member deleted.

Dependence and span identify such a member; deletion preserves the span. Finite termination gives an independent spanning sublist.

For (ii), let $(\mathbf{w}_1, \dots, \mathbf{w}_s)$ span $\mathcal{V}$. Starting from the given independent list, append $\mathbf{w}_j$ exactly when it is outside the current span. At every stage,

retained list independent, $\mathbf{w}_j \in \text{span of the retained list} \quad (1 \leq j \leq s \text{ already inspected}).$

After s steps the retained list is independent and spans the original spanning list, hence spans $\mathcal{V}$. □

Lemma 2.13 (Steinitz exchange). Suppose $\mathbf{u}_1, \dots, \mathbf{u}_r$ is independent and $\mathbf{w}_1, \dots, \mathbf{w}_s$ spans the same vector space. Then $r \leq s$. Moreover, r members of the spanning list can be replaced by $\mathbf{u}_1, \dots, \mathbf{u}_r$ while the resulting list still spans the space.

Origin: H Thm. 4.3.2; A&M Ex. 3.28

Proof. The case $r = 0$ is immediate. At step $k \geq 1$, suppose

$$(\mathbf{u}_1, \dots, \mathbf{u}_{k-1}, \mathbf{w}_{i_k}, \dots, \mathbf{w}_{i_s})$$

spans the space. Write $\mathbf{u}_k$ as a combination of this list. At least one coefficient of an unreplaced $\mathbf{w}_{i_j}$ is nonzero; otherwise $\mathbf{u}_k \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}\}$. Solving for that $\mathbf{w}_{i_j}$ gives

$$\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{k-1}, \mathbf{w}_{i_k}, \dots, \mathbf{w}_{i_s}\} = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_{i_k}, \dots, \widehat{\mathbf{w}_{i_j}}, \dots, \mathbf{w}_{i_s}\}.$$

Each step consumes one distinct $\mathbf{w}$, so $r \leq s$; after step r the list has the asserted form. $\square$

Theorem 2.14 (All bases have the same size). Any two finite bases of a vector space contain the same number of vectors.

Origin: H Thm. 4.3.6; A&M Ex. 3.29

Proof. For bases of sizes r and s , Steinitz exchange in the two directions gives

$$r \leq s, \quad s \leq r, \quad r = s.$$

□

Definition 2.15 (Dimension). The **dimension** of a finite-dimensional vector space $\mathcal{V}$, written $\dim \mathcal{V}$, is the common number of vectors in any basis of $\mathcal{V}$. In particular, $\dim\{\mathbf{0}\} = 0$.

Origin: H §4.3e

Theorem 2.16 (Dimension of a subspace). If $\mathcal{S}$ is a subspace of a finite-dimensional space $\mathcal{V}$, then $\mathcal{S}$ has a finite basis and

$$\dim \mathcal{S} \leq \dim \mathcal{V}.$$

Equality holds if and only if $\mathcal{S} = \mathcal{V}$.

Origin: H Thms. 4.3.8 and 4.3.10

Proof. Starting with the empty list, append a vector of $\mathcal{S}$ outside the current span. Appending preserves independence, while Steinitz exchange bounds every resulting list by $\dim \mathcal{V}$. The process

stops after at most $\dim \mathcal{V}$ steps with a basis of $\mathcal{S}$; hence $\dim \mathcal{S} \leq \dim \mathcal{V}$.

The implication $\mathcal{S} = \mathcal{V} \Rightarrow \dim \mathcal{S} = \dim \mathcal{V}$ is immediate. Conversely, extend a basis $\mathcal{B}_{\mathcal{S}}$ of $\mathcal{S}$ to a basis of $\mathcal{V}$. If $\dim \mathcal{S} = \dim \mathcal{V}$, then

$$\#\mathcal{B}_{\mathcal{S}} = \dim \mathcal{S} = \dim \mathcal{V},$$

so no vector is appended and $\text{span}(\mathcal{B}_{\mathcal{S}}) = \mathcal{V}$. Thus $\mathcal{S} = \mathcal{V}$. □

Corollary 2.17 (n vectors in an n -dimensional space). In an n -dimensional vector space, a list of n vectors is linearly independent if and only if it spans the space. Either condition makes it a basis.

Origin: H Thms. 4.3.2, 4.3.7, and 4.3.9

Proof. For a list $\mathcal{L}$ of length n ,

$$\mathcal{L} \text{ independent} \implies \mathcal{L} \subseteq \mathcal{B}, \quad \#\mathcal{B} = n \implies \mathcal{L} = \mathcal{B},$$

and

$$\mathcal{L} \text{ spanning} \implies \mathcal{B} \subseteq \mathcal{L}, \quad \#\mathcal{B} = n \implies \mathcal{L} = \mathcal{B}. \quad \square$$

Example 2.6 (Three standard dimensions). The standard vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ form a basis of $\mathbb{R}^n$, so $\dim \mathbb{R}^n = n$. In $\mathbb{R}^{m \times n}$, let $\mathbf{E}_{ij}$ have a 1 in position (i, j) and zeros elsewhere. The mn matrices $\mathbf{E}_{ij}$ form a basis, so $\dim \mathbb{R}^{m \times n} = mn$. Finally, $1, t, \dots, t^d$ form a basis of $\mathcal{P}_d$, so $\dim \mathcal{P}_d = d + 1$. Spanning follows by the definitions of the three spaces, and independence follows by comparing coordinates, entries, or polynomial coefficients.

Origin: H §4.3b and Eq. (4.3.2); S §1.1.1

2.2 Coordinates and change of basis

Theorem 2.18 (Unique representation in a basis). If $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ is a basis of $\mathcal{V}$, then every $\mathbf{x} \in \mathcal{V}$ has exactly one representation

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n.$$

Origin: H Lemma 4.3.5

Proof. Spanning gives existence. Independence gives

$$\sum_{j=1}^n \alpha_j \mathbf{v}_j = \sum_{j=1}^n \beta_j \mathbf{v}_j \iff \sum_{j=1}^n (\alpha_j - \beta_j) \mathbf{v}_j = \mathbf{0} \iff (\forall j) \alpha_j = \beta_j.$$

□

Definition 2.19 (Coordinates). For an ordered basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, the **coordinate vector** of $\mathbf{x} = \sum_j \alpha_j \mathbf{v}_j$ in $\mathcal{B}$ is

$$[\mathbf{x}]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{R}^n.$$

The order of the basis vectors determines the order of the coordinates.

Origin: H §22.3; S §1.2.2

Definition 2.20 (Transition matrix). Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $\mathcal{C} = (\mathbf{w}_1, \dots, \mathbf{w}_n)$ be ordered

bases of the same space. The **transition matrix from $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates** is

$$\mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} = ([\mathbf{v}_1]_{\mathcal{C}}, \dots, [\mathbf{v}_n]_{\mathcal{C}}).$$

Its j th column records the $\mathcal{C}$ -coordinates of the j th vector of $\mathcal{B}$.

Origin: H §22.4b(3)

Theorem 2.21 (Changing coordinates). For every $\mathbf{x} \in \mathcal{V}$,

$$[\mathbf{x}]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{x}]_{\mathcal{B}}.$$

The transition matrices in the two directions satisfy

$$\mathbf{P}_{\mathcal{B} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} = \mathbf{I}_n, \quad \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} \mathbf{P}_{\mathcal{B} \leftarrow \mathcal{C}} = \mathbf{I}_n.$$

Origin: H §22.4b(3)

Proof. Write $[\mathbf{x}]_{\mathcal{B}} = (\alpha_1, \dots, \alpha_n)'$. Then

$$[\mathbf{x}]_{\mathcal{C}} = \left[\sum_{j=1}^n \alpha_j \mathbf{v}_j \right]_{\mathcal{C}} = \sum_j \alpha_j [\mathbf{v}_j]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{x}]_{\mathcal{B}}.$$

For every $\mathbf{y} = [\mathbf{x}]_{\mathcal{B}} \in \mathbb{R}^n$,

$$\mathbf{P}_{\mathcal{B} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} \mathbf{y} = \mathbf{y}.$$

Taking $\mathbf{y} = \mathbf{e}_j$ for $j = 1, \dots, n$ gives

$$\mathbf{P}_{\mathcal{B} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} = \mathbf{I}_n.$$

Interchanging $\mathcal{B}$ and $\mathcal{C}$ gives the second identity. □

Example 2.7 (Two coordinate systems). Let $\mathcal{E} = (\mathbf{e}_1, \mathbf{e}_2)$ be the standard basis of $\mathbb{R}^2$ and

$$\mathcal{B} = \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right).$$

For $\mathbf{x} = (5, 1)'$, the identity

$$\mathbf{x} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

gives $[\mathbf{x}]_{\mathcal{B}} = (3, 2)'$, whereas $[\mathbf{x}]_{\mathcal{E}} = (5, 1)'$. The transition matrices are

$$\mathbf{P}_{\mathcal{E} \leftarrow \mathcal{B}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{P}_{\mathcal{B} \leftarrow \mathcal{E}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Their products in both orders equal $\mathbf{I}_2$, and $\mathbf{P}_{\mathcal{E} \leftarrow \mathcal{B}}(3, 2)' = (5, 1)'$.

Origin: Course formulation from H §22.4b(3)

2.3 Sums, direct sums, and the dimension formula

Definition 2.22 (Sum and direct sum). For subspaces $\mathcal{S}, \mathcal{T} \subseteq \mathcal{V}$, define

$$\mathcal{S} + \mathcal{T} = \{\mathbf{s} + \mathbf{t} : \mathbf{s} \in \mathcal{S}, \mathbf{t} \in \mathcal{T}\}.$$

The sum is **direct**, written $\mathcal{S} \oplus \mathcal{T}$, if each member of $\mathcal{S} + \mathcal{T}$ has a unique expression $\mathbf{s} + \mathbf{t}$ with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$.

Origin: S §1.1.1; A&M Ex. 3.12(c)

Theorem 2.23 (The sum is a subspace). The set $\mathcal{S} + \mathcal{T}$ is the smallest subspace of $\mathcal{V}$ containing both $\mathcal{S}$ and $\mathcal{T}$.

Origin: S §1.1.1; A&M Ex. 3.12(c)

Proof.

$$\mathbf{0} = \mathbf{0} + \mathbf{0} \in \mathcal{S} + \mathcal{T}.$$

For $\mathbf{s}_i \in \mathcal{S}$ and $\mathbf{t}_i \in \mathcal{T}$,

$$\alpha(\mathbf{s}_1 + \mathbf{t}_1) + \beta(\mathbf{s}_2 + \mathbf{t}_2) = (\alpha\mathbf{s}_1 + \beta\mathbf{s}_2) + (\alpha\mathbf{t}_1 + \beta\mathbf{t}_2),$$

where the terms on the right belong to $\mathcal{S}$ and $\mathcal{T}$. Moreover,

$$\mathcal{S} \subseteq \mathcal{S} + \mathcal{T}, \quad \mathcal{T} \subseteq \mathcal{S} + \mathcal{T}.$$

If $\mathcal{U}$ is a subspace containing $\mathcal{S} \cup \mathcal{T}$, then

$$\mathbf{s} \in \mathcal{S}, \mathbf{t} \in \mathcal{T} \implies \mathbf{s} + \mathbf{t} \in \mathcal{U} \implies \mathcal{S} + \mathcal{T} \subseteq \mathcal{U}.$$



Theorem 2.24 (Characterisations of a direct sum). For subspaces $\mathcal{S}$ and $\mathcal{T}$, the following are equivalent:

- (i) the sum $\mathcal{S} + \mathcal{T}$ is direct;
- (ii) $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$;
- (iii) $\mathbf{s} + \mathbf{t} = \mathbf{0}$, with $\mathbf{s} \in \mathcal{S}$ and $\mathbf{t} \in \mathcal{T}$, implies $\mathbf{s} = \mathbf{t} = \mathbf{0}$.

Origin: A&M Ex. 3.13

Proof.

- (i) $\implies \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v}, \quad \mathbf{v} \in \mathcal{S} \cap \mathcal{T} \implies \mathbf{v} = \mathbf{0},$
- (ii) $\implies \mathbf{s} + \mathbf{t} = \mathbf{0} \implies \mathbf{s} = -\mathbf{t} \in \mathcal{S} \cap \mathcal{T} \implies \mathbf{s} = \mathbf{t} = \mathbf{0},$
- (iii) $\implies \mathbf{s}_1 + \mathbf{t}_1 = \mathbf{s}_2 + \mathbf{t}_2 \implies (\mathbf{s}_1 - \mathbf{s}_2) + (\mathbf{t}_1 - \mathbf{t}_2) = \mathbf{0} \implies \mathbf{s}_1 = \mathbf{s}_2, \mathbf{t}_1 = \mathbf{t}_2.$

□

Theorem 2.25 (Dimension formula). If $\mathcal{S}$ and $\mathcal{T}$ are finite-dimensional subspaces, then

$$\dim(\mathcal{S} + \mathcal{T}) + \dim(\mathcal{S} \cap \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T}.$$

Origin: H Thm. 17.4.1; S Prop. 1.2

Proof. Let $\mathbf{u}_1, \dots, \mathbf{u}_q$ be a basis of $\mathcal{S} \cap \mathcal{T}$. Extend it separately to bases

$$(\mathbf{u}_1, \dots, \mathbf{u}_q, \mathbf{s}_1, \dots, \mathbf{s}_r) \quad \text{of } \mathcal{S},$$

and

$$(\mathbf{u}_1, \dots, \mathbf{u}_q, \mathbf{t}_1, \dots, \mathbf{t}_\ell) \quad \text{of } \mathcal{T}.$$

The list

$$\mathcal{B} = (\mathbf{u}_1, \dots, \mathbf{u}_q, \mathbf{s}_1, \dots, \mathbf{s}_r, \mathbf{t}_1, \dots, \mathbf{t}_\ell)$$

spans $\mathcal{S} + \mathcal{T}$. For a zero combination,

$$\sum_i a_i \mathbf{u}_i + \sum_j b_j \mathbf{s}_j + \sum_k c_k \mathbf{t}_k = \mathbf{0}.$$

Then

$$\sum_k c_k \mathbf{t}_k = - \sum_i a_i \mathbf{u}_i - \sum_j b_j \mathbf{s}_j \in \mathcal{S} \cap \mathcal{T} = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_q\}.$$

Independence of the basis of $\mathcal{T}$ gives $c_k = 0$ for all k ; independence of the basis of $\mathcal{S}$ gives $a_i = b_j = 0$ for all i, j . Thus $\mathcal{B}$ is a basis, and

$$\dim(\mathcal{S} + \mathcal{T}) = q + r + \ell, \quad \dim \mathcal{S} = q + r, \quad \dim \mathcal{T} = q + \ell, \quad \dim(\mathcal{S} \cap \mathcal{T}) = q.$$

□

Corollary 2.26 (Dimension of a direct sum). If $\mathcal{S}$ and $\mathcal{T}$ are finite-dimensional and $\mathcal{S} + \mathcal{T}$ is direct, then

$$\dim(\mathcal{S} \oplus \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T}.$$

Origin: H Cor. 17.1.6; S Prop. 1.2

Proof.

$$\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\} \implies \dim(\mathcal{S} \oplus \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim\{\mathbf{0}\} = \dim \mathcal{S} + \dim \mathcal{T}.$$

□

Example 2.8 (Two planes in $\mathbb{R}^3$). Let $\mathcal{S} = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\mathcal{T} = \text{span}\{\mathbf{e}_2, \mathbf{e}_3\}$. Then

$$\mathcal{S} \cap \mathcal{T} = \text{span}\{\mathbf{e}_2\}, \quad \mathcal{S} + \mathcal{T} = \mathbb{R}^3.$$

The dimension formula reads $3 + 1 = 2 + 2$. The sum is not direct. In contrast, $\text{span}\{\mathbf{e}_1, \mathbf{e}_2\} + \text{span}\{\mathbf{e}_3\}$ is a direct sum equal to $\mathbb{R}^3$.

Origin: Course formulation from H Thm. 17.4.1

3 The subspaces of a matrix

3.1 Column space, row space, null space

Definition 2.27 (Linear map, image, kernel). Let $\mathcal{V}$ and $\mathcal{W}$ be real vector spaces. A map $T : \mathcal{V} \rightarrow \mathcal{W}$ is **linear** if

$$T(\alpha \mathbf{u} + \beta \mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$$

for every $\mathbf{u}, \mathbf{v} \in \mathcal{V}$ and $\alpha, \beta \in \mathbb{R}$. Its **image** and **kernel** are

$$\text{im}(T) = \{T(\mathbf{v}) : \mathbf{v} \in \mathcal{V}\}, \quad \ker(T) = \{\mathbf{v} \in \mathcal{V} : T(\mathbf{v}) = \mathbf{0}\}.$$

The map is **one-to-one** if $T(\mathbf{u}) = T(\mathbf{v})$ implies $\mathbf{u} = \mathbf{v}$, and **onto** if $\text{im}(T) = \mathcal{W}$.

Origin: H §22.1

Theorem 2.28 (Image and kernel are subspaces). For a linear map $T : \mathcal{V} \rightarrow \mathcal{W}$, the image is a subspace of $\mathcal{W}$ and the kernel is a subspace of $\mathcal{V}$.

Origin: H §22.1

Proof.

$$T(\mathbf{0}) = T(0\mathbf{u}) = 0T(\mathbf{u}) = \mathbf{0}.$$

For $\mathbf{w}_i = T(\mathbf{v}_i) \in \text{im}(T)$,

$$\alpha\mathbf{w}_1 + \beta\mathbf{w}_2 = T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2),$$

so $\alpha\mathbf{w}_1 + \beta\mathbf{w}_2 \in \text{im}(T)$. For $\mathbf{v}_i \in \ker(T)$,

$$T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha\mathbf{0} + \beta\mathbf{0} = \mathbf{0},$$

so $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 \in \ker(T)$. □

Definition 2.29 (Three matrix subspaces). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and rows $\tilde{\mathbf{a}}'_1, \dots, \tilde{\mathbf{a}}'_m$. Its **column space**, **row space**, and **null space** are

$$\begin{aligned}\operatorname{col}(\mathbf{A}) &= \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} \subseteq \mathbb{R}^m, \\ \operatorname{row}(\mathbf{A}) &= \operatorname{span}\{\tilde{\mathbf{a}}'_1, \dots, \tilde{\mathbf{a}}'_m\} = \operatorname{col}(\mathbf{A}') \subseteq \mathbb{R}^n, \\ \operatorname{null}(\mathbf{A}) &= \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n.\end{aligned}$$

Origin: H §§4.1a, 4.1c, and 11.1

Proposition 2.30 (The matrix map). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, the map $T_{\mathbf{A}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ is linear, with

$$\operatorname{im}(T_{\mathbf{A}}) = \operatorname{col}(\mathbf{A}), \quad \operatorname{ker}(T_{\mathbf{A}}) = \operatorname{null}(\mathbf{A}).$$

Consequently, all three sets in Definition 2.29 are subspaces of their stated ambient spaces.

Origin: H Thm. 22.5.1; S §2.2.1

Proof.

$$T_{\mathbf{A}}(\alpha \mathbf{x} + \beta \mathbf{y}) = \mathbf{A}(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha \mathbf{A}\mathbf{x} + \beta \mathbf{A}\mathbf{y}.$$

For $\mathbf{x} = (x_1, \dots, x_n)'$,

$$T_{\mathbf{A}}(\mathbf{x}) = \mathbf{A}\mathbf{x} = \sum_{j=1}^n x_j \mathbf{a}_j \implies \text{im}(T_{\mathbf{A}}) = \text{col}(\mathbf{A}).$$

$$\ker(T_{\mathbf{A}}) = \{\mathbf{x} : \mathbf{A}\mathbf{x} = \mathbf{0}\} = \text{null}(\mathbf{A}).$$

The image–kernel result applied to $\mathbf{A}$ and $\mathbf{A}'$ gives the subspace assertions. □

Theorem 2.31 (Subspace containment and factorisation). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{m \times p}$, and $\mathbf{C} \in \mathbb{R}^{q \times n}$. Then

$$\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A}) \iff \mathbf{B} = \mathbf{A}\mathbf{F} \quad \text{for some } \mathbf{F} \in \mathbb{R}^{n \times p},$$

and

$$\text{row}(\mathbf{C}) \subseteq \text{row}(\mathbf{A}) \iff \mathbf{C} = \mathbf{L}\mathbf{A} \quad \text{for some } \mathbf{L} \in \mathbb{R}^{q \times m}.$$

Origin: H Lemma 4.2.2

Proof. Write $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_p)$. Then

$$\begin{aligned} \text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A}) &\iff (\forall j)(\exists \mathbf{f}_j \in \mathbb{R}^n) \mathbf{b}_j = \mathbf{A}\mathbf{f}_j \\ &\iff \mathbf{B} = \mathbf{A}(\mathbf{f}_1, \dots, \mathbf{f}_p). \end{aligned}$$

If $\mathbf{B} = \mathbf{A}\mathbf{F}$, then $\mathbf{B}\mathbf{u} = \mathbf{A}(\mathbf{F}\mathbf{u}) \in \text{col}(\mathbf{A})$, proving the reverse implication. The row statement is the column statement applied to $\mathbf{C}'$ and $\mathbf{A}'$:

$$\text{row}(\mathbf{C}) \subseteq \text{row}(\mathbf{A}) \iff \mathbf{C}' = \mathbf{A}'\mathbf{L}' \iff \mathbf{C} = \mathbf{L}\mathbf{A}.$$

□

Corollary 2.32 (Consistency). The system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b} \in \text{col}(\mathbf{A})$.

Origin: H Thm. 7.2.1

Proof.

$$\mathbf{Ax} = \mathbf{b} \text{ is consistent} \iff (\exists \mathbf{x} \in \mathbb{R}^n) \mathbf{b} = T_{\mathbf{A}}(\mathbf{x}) \iff \mathbf{b} \in \text{im}(T_{\mathbf{A}}) \iff \mathbf{b} \in \text{col}(\mathbf{A}).$$

□

Theorem 2.33 (All solutions of a linear system). If $\mathbf{x}_0$ is one solution of $\mathbf{Ax} = \mathbf{b}$, then the set of all solutions is

$$\mathbf{x}_0 + \text{null}(\mathbf{A}) = \{\mathbf{x}_0 + \mathbf{z} : \mathbf{z} \in \text{null}(\mathbf{A})\}.$$

The solution is unique if and only if $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$. If the null space contains a nonzero vector, every consistent system has infinitely many solutions.

Origin: H Thm. 11.2.3

Proof.

$$\mathbf{Ax} = \mathbf{b} \iff \mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0} \iff \mathbf{x} - \mathbf{x}_0 \in \text{null}(\mathbf{A}) \iff \mathbf{x} \in \mathbf{x}_0 + \text{null}(\mathbf{A}).$$

Hence uniqueness is equivalent to $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$. If $\mathbf{0} \neq \mathbf{z} \in \text{null}(\mathbf{A})$, then

$$\mathbf{A}(\mathbf{x}_0 + t\mathbf{z}) = \mathbf{b}, \quad \mathbf{x}_0 + t_1\mathbf{z} = \mathbf{x}_0 + t_2\mathbf{z} \iff (t_1 - t_2)\mathbf{z} = \mathbf{0} \iff t_1 = t_2.$$

□

Example 2.9 (Three subspaces of one matrix). Consider

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix}.$$

Its third column is the sum of its first two and its fourth is twice the first minus the second. The first two columns are independent, so

$$\text{col}(\mathbf{A}) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : z = x + y \right\}.$$

The third row is the sum of the first two, which are independent, so

$$\text{row}(\mathbf{A}) = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} \right\}.$$

The equations $\mathbf{A}\mathbf{x} = \mathbf{0}$ reduce to

$$x_1 + x_3 + 2x_4 = 0, \quad x_2 + x_3 - x_4 = 0.$$

With $x_3 = s$ and $x_4 = t$, they give

$$\text{null}(\mathbf{A}) = \text{span} \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

The right-hand side $(1, 2, 3)'$ is reachable because its third component is the sum of its first two, and all solutions are

$$\begin{pmatrix} 1 \\ 2 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

The right-hand side $(1, 2, 4)'$ is not in the column space, so its system has no solution.

Origin: Course formulation from H §§4.1 and 11.1

3.2 Rank and nullity

Definition 2.34 (Rank and nullity). For $\mathbf{A} \in \mathbb{R}^{m \times n}$, define

$$\text{rank}(\mathbf{A}) = \dim \text{col}(\mathbf{A}), \quad \text{nullity}(\mathbf{A}) = \dim \text{null}(\mathbf{A}).$$

Its **row rank** is $\dim \text{row}(\mathbf{A})$. The matrix has **full column rank** if $\text{rank}(\mathbf{A}) = n$ and **full row rank** if $\text{rank}(\mathbf{A}) = m$.

Origin: H §§4.4a and 11.1

Theorem 2.35 (Rank–nullity). If $T : \mathcal{V} \rightarrow \mathcal{W}$ is linear and $\mathcal{V}$ is finite-dimensional, then

$$\dim \ker(T) + \dim \operatorname{im}(T) = \dim \mathcal{V}.$$

In particular, if $\mathbf{A} \in \mathbb{R}^{m \times n}$, then

$$\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = n.$$

Origin: H Thm. 22.1.1

Proof. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be a basis of $\ker(T)$ and extend it to a basis

$$\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n$$

of $\mathcal{V}$. For $\mathbf{x} = \sum_{j=1}^n a_j \mathbf{v}_j$,

$$T(\mathbf{x}) = \sum_{j=1}^n a_j T(\mathbf{v}_j) = \sum_{j=k+1}^n a_j T(\mathbf{v}_j),$$

so $T(\mathbf{v}_{k+1}), \dots, T(\mathbf{v}_n)$ spans $\text{im}(T)$. Moreover,

$$\sum_{j=k+1}^n a_j T(\mathbf{v}_j) = \mathbf{0} \implies \sum_{j=k+1}^n a_j \mathbf{v}_j \in \ker(T) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \implies a_{k+1} = \dots = a_n = 0.$$

Thus these $n - k$ images form a basis of $\text{im}(T)$, and

$$\dim \ker(T) + \dim \text{im}(T) = k + (n - k) = n.$$

For $T = T_{\mathbf{A}}$, image and kernel are $\text{col}(\mathbf{A})$ and $\text{null}(\mathbf{A})$. □

Theorem 2.36 (Row rank equals column rank). For every matrix $\mathbf{A}$, $\dim \text{row}(\mathbf{A}) = \dim \text{col}(\mathbf{A}) = \text{rank}(\mathbf{A})$.

Origin: H Thms. 4.4.1–4.4.2; S Prop. 2.7

Proof. Let $r = \dim \text{col}(\mathbf{A})$. For $r = 0$, $\mathbf{A} = \mathbf{O}$ and both dimensions are zero. For $r > 0$, put r basis columns of $\mathbf{A}$ in $\mathbf{B} \in \mathbb{R}^{m \times r}$. Their coordinate vectors give $\mathbf{C} \in \mathbb{R}^{r \times n}$ such that $\mathbf{A} = \mathbf{BC}$. Hence

$$\text{row}(\mathbf{A}) \subseteq \text{row}(\mathbf{C}), \quad \dim \text{row}(\mathbf{A}) \leq r = \dim \text{col}(\mathbf{A}).$$

Applying the inequality to $\mathbf{A}'$ gives

$$\dim \text{col}(\mathbf{A}) = \dim \text{row}(\mathbf{A}') \leq \dim \text{col}(\mathbf{A}') = \dim \text{row}(\mathbf{A}).$$

□

Corollary 2.37 (Basic rank facts). For $\mathbf{A} \in \mathbb{R}^{m \times n}$:

- (i) $\text{rank}(\mathbf{A}') = \text{rank}(\mathbf{A}) \leq \min\{m, n\}$;
- (ii) $\mathbf{A}$ has full column rank if and only if $\text{null}(\mathbf{A}) = \{\mathbf{0}\}$, equivalently, $T_{\mathbf{A}}$ is one-to-one;

(iii) $\mathbf{A}$ has full row rank if and only if $\text{col}(\mathbf{A}) = \mathbb{R}^m$, equivalently, every right-hand side in $\mathbb{R}^m$ is reachable.

Origin: H Lemma 4.4.3, §4.4c, and Lemma 22.1.3

Proof.

$$\text{rank}(\mathbf{A}') = \dim \text{row}(\mathbf{A}) = \text{rank}(\mathbf{A}), \quad \text{rank}(\mathbf{A}) \leq \min\{m, n\}.$$

Rank-nullity and the matrix-map identities give

$$\begin{aligned} \text{rank}(\mathbf{A}) = n &\iff \dim \text{null}(\mathbf{A}) = 0 \iff \text{null}(\mathbf{A}) = \{\mathbf{0}\} \\ &\iff (\mathbf{Ax} = \mathbf{Ay} \implies \mathbf{x} - \mathbf{y} = \mathbf{0}) \iff T_{\mathbf{A}} \text{ is one-to-one.} \end{aligned}$$

Finally,

$$\text{rank}(\mathbf{A}) = m \iff \dim \text{col}(\mathbf{A}) = \dim \mathbb{R}^m \iff \text{col}(\mathbf{A}) = \mathbb{R}^m \iff (\forall \mathbf{b} \in \mathbb{R}^m)(\exists \mathbf{x}) \mathbf{Ax} = \mathbf{b}.$$

□

Example 2.10 (Rank and nullity of the running matrix). The three bases in Example 2.9 have respectively two, two, and two members. Thus the matrix has column rank 2, row rank 2, and nullity 2. Rank–nullity reads $2 + 2 = 4$, the number of columns.

Origin: Course formulation from H Thm. 22.1.1

3.3 The model space

Definition 2.38 (Model space and identifiability). For a model matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, the **model space** is

$$\mathcal{M}(\mathbf{X}) = \text{col}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\}.$$

It is the set of linear-predictor vectors that the model can represent. The coefficient parameterisation is **identifiable** if $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\gamma}$ implies $\boldsymbol{\beta} = \boldsymbol{\gamma}$.

Origin: Course formulation from H §11.1 and Lemma 22.1.3

Corollary 2.39 (Equivalent model matrices). For $\mathbf{X} \in \mathbb{R}^{n \times p}$ and $\mathbf{Z} \in \mathbb{R}^{n \times q}$,

$$\mathcal{M}(\mathbf{Z}) \subseteq \mathcal{M}(\mathbf{X}) \iff \mathbf{Z} = \mathbf{X}\mathbf{F} \quad \text{for some } \mathbf{F} \in \mathbb{R}^{p \times q}.$$

Consequently,

$$\mathcal{M}(\mathbf{Z}) = \mathcal{M}(\mathbf{X}) \iff \mathbf{Z} = \mathbf{X}\mathbf{F} \text{ and } \mathbf{X} = \mathbf{Z}\mathbf{G} \text{ for some } \mathbf{F}, \mathbf{G}.$$

Origin: Course formulation from H Lemma 4.2.2

Proof.

$$\mathcal{M}(\mathbf{Z}) \subseteq \mathcal{M}(\mathbf{X}) \iff \text{col}(\mathbf{Z}) \subseteq \text{col}(\mathbf{X}) \iff (\exists \mathbf{F}) \mathbf{Z} = \mathbf{X}\mathbf{F}.$$

Equality is mutual inclusion. □

Theorem 2.40 (Coefficient equivalence). For any $\boldsymbol{\beta}, \boldsymbol{\gamma} \in \mathbb{R}^p$,

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\gamma} \iff \boldsymbol{\beta} - \boldsymbol{\gamma} \in \text{null}(\mathbf{X}).$$

Consequently, the coefficient parameterisation is identifiable if and only if $\mathbf{X}$ has full column rank.

If $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}_0$, then

$$\{\boldsymbol{\beta} \in \mathbb{R}^p : \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\eta}\} = \boldsymbol{\beta}_0 + \text{null}(\mathbf{X}).$$

Origin: Course formulation from H Thm. 11.2.3 and Lemma 22.1.3

Proof.

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\gamma} \iff \mathbf{X}(\boldsymbol{\beta} - \boldsymbol{\gamma}) = \mathbf{0} \iff \boldsymbol{\beta} - \boldsymbol{\gamma} \in \text{null}(\mathbf{X}).$$

Therefore

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\gamma} \implies \boldsymbol{\beta} = \boldsymbol{\gamma} \iff \text{null}(\mathbf{X}) = \{\mathbf{0}\} \iff \text{rank}(\mathbf{X}) = p.$$

For $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}_0$,

$$\{\boldsymbol{\beta} : \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\eta}\} = \boldsymbol{\beta}_0 + \text{null}(\mathbf{X}).$$

□

Example 2.11 (Factor codings and the dummy-variable trap). Suppose a factor has c levels, every level occurs, and $\mathbf{d}_j \in \mathbb{R}^n$ is the indicator vector of level j . Each observation belongs to exactly one level, so

$$\mathbf{d}_1 + \cdots + \mathbf{d}_c = \mathbf{1}.$$

The indicator vectors are independent: if $\sum_j a_j \mathbf{d}_j = \mathbf{0}$, inspect an observation in level j to obtain $a_j = 0$. Thus $(\mathbf{d}_1, \dots, \mathbf{d}_c)$ is a basis of the factor's model space.

Including an intercept and all c indicators gives

$$\mathbf{X}_F = (\mathbf{1}, \mathbf{d}_1, \dots, \mathbf{d}_c), \quad \mathbf{X}_F \begin{pmatrix} 1 \\ -1 \\ \vdots \\ -1 \end{pmatrix} = \mathbf{0}.$$

This $c+1$ column list is redundant and has rank c . Two nonredundant choices are the **cell-means coding**

$$\mathbf{X}_C = (\mathbf{d}_1, \dots, \mathbf{d}_c)$$

and the **treatment coding**, with level 1 as baseline,

$$\mathbf{X}_T = (\mathbf{1}, \mathbf{d}_2, \dots, \mathbf{d}_c).$$

The treatment list is independent: an equation $a_0\mathbf{1} + \sum_{j=2}^c a_j\mathbf{d}_j = \mathbf{0}$ gives $a_0 = 0$ on level 1 and then $a_j = 0$ on level j . It spans the same model space because $\mathbf{1} = \sum_j \mathbf{d}_j$ and $\mathbf{d}_1 = \mathbf{1} - \sum_{j=2}^c \mathbf{d}_j$. The two column lists are therefore bases of one model space.

For three levels, cell means $(\mu_1, \mu_2, \mu_3)'$ correspond to treatment coefficients

$$\begin{pmatrix} \alpha_0 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 - \mu_1 \\ \mu_3 - \mu_1 \end{pmatrix}.$$

Both codings produce value μ_j at every observation in level j . The redundant coding is the dummy-variable trap; either coding removes it without changing the model space.

Origin: Course formulation from H Lemma 4.2.2 and Thm. 11.2.3

4 Computation: Numerical Checks in R

4.1 Column, row, and null-space identities

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 2 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{K} = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \end{pmatrix},$$

$$\mathbf{N} = \begin{pmatrix} -1 & -2 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{A} = \mathbf{BK}, \quad \tilde{\mathbf{a}}'_3 = \tilde{\mathbf{a}}'_1 + \tilde{\mathbf{a}}'_2, \quad \mathbf{AN} = \mathbf{O}.$$

```

1 A <- rbind(c(1, 0, 1, 2),
2           c(0, 1, 1, -1),
3           c(1, 1, 2, 1))
4 B <- A[, 1:2, drop = FALSE]
5 K <- rbind(c(1, 0, 1, 2),
6           c(0, 1, 1, -1))
7 Nhand <- cbind(c(-1, -1, 1, 0),
8               c(-2, 1, 0, 1))
9 c(A_minus_BK = max(abs(A - B %*% K)),
10   row3_minus_row1_row2 = max(abs(A[3, ] - A[1, ] - A[2, ])),
11   AN = max(abs(A %*% Nhand)))

```

```

##           A_minus_BK row3_minus_row1_row2
##                0                0
##                AN

```

##

0

4.2 Coordinate identities

$$P_{\mathcal{E} \leftarrow \mathcal{B}} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad P_{\mathcal{B} \leftarrow \mathcal{E}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

$$P_{\mathcal{E} \leftarrow \mathcal{B}} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad P_{\mathcal{B} \leftarrow \mathcal{E}} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix},$$

$$P_{\mathcal{B} \leftarrow \mathcal{E}} P_{\mathcal{E} \leftarrow \mathcal{B}} = P_{\mathcal{E} \leftarrow \mathcal{B}} P_{\mathcal{B} \leftarrow \mathcal{E}} = \mathbf{I}_2.$$

```

1 P_E_from_B <- matrix(c(1, 1, 1, -1), 2, 2)
2 P_B_from_E <- 0.5 * P_E_from_B
3 v_E <- c(5, 1)
4 v_B <- c(3, 2)
5 c(E_coordinates = max(abs(P_E_from_B %*% v_B - v_E)),
6   B_coordinates = max(abs(P_B_from_E %*% v_E - v_B)),
7   B_round_trip = max(abs(P_B_from_E %*% P_E_from_B - diag(2))),
8   E_round_trip = max(abs(P_E_from_B %*% P_B_from_E - diag(2))))

```

```
## E_coordinates B_coordinates B_round_trip E_round_trip
##              0              0              0              0
```

4.3 Rank–nullity from hand bases

$$\mathbf{B} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \mathbf{0} \implies c_1 = c_2 = 0, \quad \mathbf{N} \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \mathbf{0} \implies d_1 = d_2 = 0.$$

$$\text{col}(\mathbf{B}) \subseteq \text{col}(\mathbf{A}), \quad \mathbf{A} = \mathbf{BK} \implies \text{col}(\mathbf{A}) \subseteq \text{col}(\mathbf{B}), \quad \text{rank}(\mathbf{A}) = 2.$$

$$\mathbf{AN} = \mathbf{0}, \quad \dim \text{null}(\mathbf{A}) = 4 - 2 = 2, \quad \text{col}(\mathbf{N}) = \text{null}(\mathbf{A}).$$

```
1 c(B_identity_block = max(abs(B[1:2, ] - diag(2))),
2   N_identity_block = max(abs(Nhand[3:4, ] - diag(2))),
3   null_residual = max(abs(A %*% Nhand)),
4   number_of_columns = ncol(A),
5   rank = ncol(B),
6   nullity = ncol(Nhand),
7   rank_plus_nullity = ncol(B) + ncol(Nhand))
```

```
## B_identity_block N_identity_block null_residual
##              0              0              0
```

```
## number_of_columns      rank      nullity
##                4          2          2
## rank_plus_nullity
##                4
```

4.4 Indicator codings

For $\mathbf{D} = (\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3)$ and $\mathbf{X}_T = (\mathbf{1}, \mathbf{d}_2, \mathbf{d}_3)$,

$$\mathbf{1} = \mathbf{d}_1 + \mathbf{d}_2 + \mathbf{d}_3, \quad \mathbf{d}_1 = \mathbf{1} - \mathbf{d}_2 - \mathbf{d}_3, \quad \text{col}(\mathbf{D}) = \text{col}(\mathbf{X}_T).$$

If $\boldsymbol{\mu} = (3, 6, 11)'$ and $\boldsymbol{\alpha} = (3, 3, 8)'$, then

$$\mathbf{D}\boldsymbol{\mu} = \mathbf{X}_T\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (\mu_1, \mu_2 - \mu_1, \mu_3 - \mu_1)'.$$

```
1 D <- rbind(c(1, 0, 0), c(1, 0, 0),
2           c(0, 1, 0), c(0, 1, 0),
3           c(0, 0, 1), c(0, 0, 1))
4 one6 <- rep(1, 6)
5 X_all <- cbind(one6, D)
6 X_treatment <- cbind(one6, D[, 2], D[, 3])
7 mu <- c(3, 6, 11)
```

```

8 alpha <- c(mu[1], mu[2] - mu[1], mu[3] - mu[1])
9 c(indicator_sum = max(abs(D %*% rep(1, 3) - one6)),
10   redundant_coding = max(abs(X_all %*% c(1, -1, -1, -1))),
11   same_predictor = max(abs(D %*% mu - X_treatment %*% alpha)))

```

```

##      indicator_sum redundant_coding  same_predictor
##              0              0              0

```

5 Pitfalls

5.1 Calling every solution set a subspace

$$\mathcal{S}_{\mathbf{b}} = \{\mathbf{x} : \mathbf{A}\mathbf{x} = \mathbf{b}\} = \mathbf{x}_0 + \text{null}(\mathbf{A}), \quad \mathbf{b} \neq \mathbf{0} \implies \mathbf{0} \notin \mathcal{S}_{\mathbf{b}}.$$

For $\mathcal{S} = \{(x_1, x_2)' : x_1 + x_2 = 1\}$,

$$(1, 0)', (0, 1)' \in \mathcal{S}, \quad (1, 0)' + (0, 1)' \notin \mathcal{S}.$$

5.2 Counting generators as dimension

$$\dim \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \leq k, \quad \dim \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = k \iff \mathbf{v}_1, \dots, \mathbf{v}_k \text{ are independent.}$$

If $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$, then

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}.$$

5.3 Reading "the" basis of a subspace as unique

If $(\mathbf{u}, \mathbf{v})$ is a basis, then so is $(\mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v})$ because

$$\mathbf{u} = \frac{1}{2}[(\mathbf{u} + \mathbf{v}) + (\mathbf{u} - \mathbf{v})], \quad \mathbf{v} = \frac{1}{2}[(\mathbf{u} + \mathbf{v}) - (\mathbf{u} - \mathbf{v})].$$

Every basis of the same space has the same cardinality; its vectors, scales, signs, and order need not agree.

6 Exercises

Exercise 2.1. Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be linearly independent vectors. Prove that

$$\mathbf{x} + \mathbf{y}, \quad \mathbf{x} + \mathbf{z}, \quad \mathbf{y} + \mathbf{z}$$

are linearly independent.

Origin: H Ex. 3.2; A&M Ex. 3.17(a)

Solution. If

$$a(\mathbf{x} + \mathbf{y}) + b(\mathbf{x} + \mathbf{z}) + c(\mathbf{y} + \mathbf{z}) = \mathbf{0},$$

then

$$(a + b)\mathbf{x} + (a + c)\mathbf{y} + (b + c)\mathbf{z} = \mathbf{0}.$$

Independence gives

$$a + b = 0, \quad a + c = 0, \quad b + c = 0.$$

Consequently,

$$2a = (a + b) + (a + c) - (b + c) = 0,$$

and similarly $2b = 2c = 0$. Thus $a = b = c = 0$. 

Exercise 2.2. If $\mathbf{x} + \mathbf{y} + \mathbf{z} = \mathbf{0}$, prove

$$\text{span}\{\mathbf{x}, \mathbf{y}\} = \text{span}\{\mathbf{x}, \mathbf{z}\} = \text{span}\{\mathbf{y}, \mathbf{z}\}.$$

Origin: H Ex. 4.4

Solution. The identities

$$\mathbf{z} = -\mathbf{x} - \mathbf{y}, \quad \mathbf{y} = -\mathbf{x} - \mathbf{z}$$

give

$$\text{span}\{\mathbf{x}, \mathbf{z}\} \subseteq \text{span}\{\mathbf{x}, \mathbf{y}\}, \quad \text{span}\{\mathbf{x}, \mathbf{y}\} \subseteq \text{span}\{\mathbf{x}, \mathbf{z}\}.$$

Hence $\text{span}\{\mathbf{x}, \mathbf{y}\} = \text{span}\{\mathbf{x}, \mathbf{z}\}$. Cyclic permutation of $\mathbf{x}, \mathbf{y}, \mathbf{z}$ gives the remaining equality. ■

Exercise 2.3. Let $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ span $\mathcal{V}$ but be linearly dependent. Prove that every $\mathbf{x} \in \mathcal{V}$ has infinitely many distinct coefficient vectors $(a_1, \dots, a_k)'$ satisfying

$$\mathbf{x} = \sum_{i=1}^k a_i \mathbf{v}_i.$$

Origin: H Ex. 4.7

Solution. Choose $\mathbf{c} = (c_1, \dots, c_k)' \neq \mathbf{0}$ such that

$$\sum_{i=1}^k c_i \mathbf{v}_i = \mathbf{0}.$$

For $\mathbf{x} \in \mathcal{V}$, choose $\mathbf{a} = (a_1, \dots, a_k)'$ with $\mathbf{x} = \sum_i a_i \mathbf{v}_i$. Then, for every $t \in \mathbb{R}$,

$$\sum_{i=1}^k (a_i + tc_i) \mathbf{v}_i = \mathbf{x} + t\mathbf{0} = \mathbf{x}.$$

If $s \neq t$, then

$$(\mathbf{a} + s\mathbf{c}) - (\mathbf{a} + t\mathbf{c}) = (s - t)\mathbf{c} \neq \mathbf{0}.$$

Thus the coefficient vectors $\mathbf{a} + t\mathbf{c}$, $t \in \mathbb{R}$, are distinct. ■

Exercise 2.4. For $n \geq 1$, let

$$\mathcal{S}_n = \{\mathbf{A} \in \mathbb{R}^{n \times n} : \mathbf{A}' = \mathbf{A}\}, \quad \mathcal{K}_n = \{\mathbf{A} \in \mathbb{R}^{n \times n} : \mathbf{A}' = -\mathbf{A}\}.$$

Prove

$$\mathbb{R}^{n \times n} = \mathcal{S}_n \oplus \mathcal{K}_n,$$

construct bases of $\mathcal{S}_n$ and $\mathcal{K}_n$ from the matrix units $\mathbf{E}_{ij}$, and compute their dimensions.

Origin: H §4.3b; A&M Ex. 2.23

Solution. For $\mathbf{A} \in \mathbb{R}^{n \times n}$,

$$\mathbf{A} = \frac{\mathbf{A} + \mathbf{A}'}{2} + \frac{\mathbf{A} - \mathbf{A}'}{2}, \quad \frac{\mathbf{A} + \mathbf{A}'}{2} \in \mathcal{S}_n, \quad \frac{\mathbf{A} - \mathbf{A}'}{2} \in \mathcal{K}_n.$$

Moreover,

$$\mathbf{A} \in \mathcal{S}_n \cap \mathcal{K}_n \implies \mathbf{A} = \mathbf{A}' = -\mathbf{A} \implies \mathbf{A} = \mathbf{O}.$$

Hence the sum is direct. The lists

$$\mathcal{B}_\mathcal{S} = \{\mathbf{E}_{ii} : 1 \leq i \leq n\} \cup \{\mathbf{E}_{ij} + \mathbf{E}_{ji} : 1 \leq i < j \leq n\},$$

$$\mathcal{B}_\mathcal{K} = \{\mathbf{E}_{ij} - \mathbf{E}_{ji} : 1 \leq i < j \leq n\}$$

consist of symmetric and skew-symmetric matrices, respectively. If $\mathbf{A} = (a_{ij}) \in \mathcal{S}_n$, then

$$\mathbf{A} = \sum_{i=1}^n a_{ii} \mathbf{E}_{ii} + \sum_{1 \leq i < j \leq n} a_{ij} (\mathbf{E}_{ij} + \mathbf{E}_{ji}).$$

If $\mathbf{A} \in \mathcal{K}_n$, then $a_{ii} = 0$, $a_{ji} = -a_{ij}$, and

$$\mathbf{A} = \sum_{1 \leq i < j \leq n} a_{ij} (\mathbf{E}_{ij} - \mathbf{E}_{ji}).$$

In either display, comparison of entries forces every coefficient to be zero when the sum is $\mathbf{0}$. Thus the lists are bases, and

$$\dim \mathcal{S}_n = n + \binom{n}{2} = \frac{n(n+1)}{2}, \quad \dim \mathcal{K}_n = \binom{n}{2} = \frac{n(n-1)}{2}.$$



Exercise 2.5. Let $\mathcal{S}, \mathcal{T}$ be subspaces of an n -dimensional vector space, with $\dim \mathcal{S} = r$ and $\dim \mathcal{T} = s$.

(i) Prove

$$\max\{0, r + s - n\} \leq \dim(\mathcal{S} \cap \mathcal{T}) \leq \min\{r, s\}.$$

(ii) Prove

$$\mathcal{S} + \mathcal{T} = \mathcal{V} \iff \dim(\mathcal{S} \cap \mathcal{T}) = r + s - n.$$

(iii) Prove that $r + s > n$ excludes a direct sum.

Origin: H Thm. 17.4.1; S Prop. 1.2 and Cor. 1.1

Solution. The dimension formula and $\mathcal{S} + \mathcal{T} \subseteq \mathcal{V}$ give

$$\dim(\mathcal{S} \cap \mathcal{T}) = r + s - \dim(\mathcal{S} + \mathcal{T}) \geq r + s - n.$$

Dimensions are nonnegative, while

$$\mathcal{S} \cap \mathcal{T} \subseteq \mathcal{S}, \quad \mathcal{S} \cap \mathcal{T} \subseteq \mathcal{T}$$

imply

$$\dim(\mathcal{S} \cap \mathcal{T}) \leq r, \quad \dim(\mathcal{S} \cap \mathcal{T}) \leq s.$$

This proves (i). For (ii),

$$\begin{aligned}\mathcal{S} + \mathcal{T} = \mathcal{V} &\iff \dim(\mathcal{S} + \mathcal{T}) = n \\ &\iff r + s - \dim(\mathcal{S} \cap \mathcal{T}) = n.\end{aligned}$$

If $r + s > n$, part (i) gives $\dim(\mathcal{S} \cap \mathcal{T}) \geq r + s - n > 0$, whereas a direct sum requires $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$.



Exercise 2.6. For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times n}$, set

$$\mathbf{C} = \begin{pmatrix} \mathbf{A} \\ \mathbf{B} \end{pmatrix}.$$

(i) Prove $\text{null}(\mathbf{C}) = \text{null}(\mathbf{A}) \cap \text{null}(\mathbf{B})$.

(ii) Prove

$$\max\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\} \leq \text{rank}(\mathbf{C}) \leq \min\{n, \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B})\}.$$

(iii) Prove

$$\text{rank}(\mathbf{C}) = \text{rank}(\mathbf{A}) \iff \text{null}(\mathbf{A}) \subseteq \text{null}(\mathbf{B}).$$

Origin: Course formulation from H Lemmas 4.5.1 and 4.5.7, Cor. 4.5.2

Solution. For $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{C}\mathbf{x} = \mathbf{0} \iff \mathbf{A}\mathbf{x} = \mathbf{0} \text{ and } \mathbf{B}\mathbf{x} = \mathbf{0} \iff \mathbf{x} \in \text{null}(\mathbf{A}) \cap \text{null}(\mathbf{B}).$$

Write $r_A = \text{rank}(\mathbf{A})$, $r_B = \text{rank}(\mathbf{B})$, and $r_C = \text{rank}(\mathbf{C})$. Then

$$\dim \text{null}(\mathbf{A}) = n - r_A, \quad \dim \text{null}(\mathbf{B}) = n - r_B, \quad \dim \text{null}(\mathbf{C}) = n - r_C.$$

The inclusions $\text{null}(\mathbf{C}) \subseteq \text{null}(\mathbf{A}), \text{null}(\mathbf{B})$ give $r_C \geq r_A, r_B$. The dimension formula gives

$$n - r_C = \dim(\text{null}(\mathbf{A}) \cap \text{null}(\mathbf{B})) \geq \dim \text{null}(\mathbf{A}) + \dim \text{null}(\mathbf{B}) - n = (n - r_A) + (n - r_B) - n,$$

so $r_C \leq r_A + r_B$; also $r_C \leq n$. Finally,

$$\begin{aligned} r_C = r_A &\iff \dim \text{null}(\mathbf{C}) = \dim \text{null}(\mathbf{A}) \\ &\iff \text{null}(\mathbf{A}) \cap \text{null}(\mathbf{B}) = \text{null}(\mathbf{A}) \\ &\iff \text{null}(\mathbf{A}) \subseteq \text{null}(\mathbf{B}), \end{aligned}$$

where the middle equivalence uses $\text{null}(\mathbf{C}) = \text{null}(\mathbf{A}) \cap \text{null}(\mathbf{B}) \subseteq \text{null}(\mathbf{A})$. ■

Exercise 2.7. Let $\mathbf{x}_1, \dots, \mathbf{x}_k$ solve $\mathbf{A}\mathbf{x} = \mathbf{b}$. For scalars $c_1, \dots, c_k$, prove

$$\mathbf{A} \left(\sum_{i=1}^k c_i \mathbf{x}_i \right) = \left(\sum_{i=1}^k c_i \right) \mathbf{b}.$$

Deduce:

- (i) if $\sum_i c_i = 1$, then $\sum_i c_i \mathbf{x}_i$ is a solution;
- (ii) if $\mathbf{b} \neq \mathbf{0}$, the converse holds;
- (iii) if $\mathbf{b} = \mathbf{0}$, every linear combination of solutions is a solution.

Solution. Linearity gives

$$\mathbf{A} \left(\sum_{i=1}^k c_i \mathbf{x}_i \right) = \sum_{i=1}^k c_i \mathbf{A} \mathbf{x}_i = \sum_{i=1}^k c_i \mathbf{b} = \left(\sum_{i=1}^k c_i \right) \mathbf{b}.$$

If $\sum_i c_i = 1$, the right-hand side is $\mathbf{b}$. If $\mathbf{b} \neq \mathbf{0}$ and the combination is a solution, then

$$\left(\sum_i c_i - 1 \right) \mathbf{b} = \mathbf{0} \implies \sum_i c_i = 1.$$

For $\mathbf{b} = \mathbf{0}$, the displayed identity equals $\mathbf{0}$ for all $c_1, \dots, c_k$. 

Exercise 2.8. Let $\mathcal{U}$ be a subspace of a finite-dimensional vector space $\mathcal{V}$, and let $S : \mathcal{U} \rightarrow \mathcal{W}$ be linear. Construct a linear map $T : \mathcal{V} \rightarrow \mathcal{W}$ such that

$$T(\mathbf{u}) = S(\mathbf{u}), \quad \mathbf{u} \in \mathcal{U}.$$

Solution. Choose a basis $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ of $\mathcal{U}$ and extend it to a basis $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ of $\mathcal{V}$. Define

$$T\left(\sum_{i=1}^n a_i \mathbf{u}_i\right) = \sum_{i=1}^r a_i S(\mathbf{u}_i).$$

Unique basis coordinates make T well-defined. For $a, b \in \mathbb{R}$ and $\mathbf{x} = \sum_i x_i \mathbf{u}_i$, $\mathbf{y} = \sum_i y_i \mathbf{u}_i$,

$$\begin{aligned} T(a\mathbf{x} + b\mathbf{y}) &= \sum_{i=1}^r (ax_i + by_i) S(\mathbf{u}_i) \\ &= aT(\mathbf{x}) + bT(\mathbf{y}). \end{aligned}$$

For $\mathbf{u} = \sum_{i=1}^r u_i \mathbf{u}_i \in \mathcal{U}$,

$$T(\mathbf{u}) = \sum_{i=1}^r u_i S(\mathbf{u}_i) = S\left(\sum_{i=1}^r u_i \mathbf{u}_i\right) = S(\mathbf{u}).$$



Exercise 2.9. Let $T : \mathcal{V} \rightarrow \mathcal{W}$ be linear and let $\mathcal{U} \subseteq \mathcal{V}$ satisfy $\mathcal{U} \cap \ker(T) = \{\mathbf{0}\}$.

- (i) If $\mathbf{u}_1, \dots, \mathbf{u}_r$ are linearly independent in $\mathcal{U}$, prove that $T(\mathbf{u}_1), \dots, T(\mathbf{u}_r)$ are linearly independent.
- (ii) If $(\mathbf{u}_1, \dots, \mathbf{u}_r)$ is a basis of $\mathcal{U}$ and $\mathcal{V} = \mathcal{U} \oplus \ker(T)$, prove that $(T(\mathbf{u}_1), \dots, T(\mathbf{u}_r))$ is a basis of $\text{im}(T)$.

Origin: H Ex. 22.3

Solution. If $\sum_i a_i T(\mathbf{u}_i) = \mathbf{0}$, then

$$\sum_i a_i \mathbf{u}_i \in \mathcal{U} \cap \ker(T) = \{\mathbf{0}\}.$$

Independence of $\mathbf{u}_1, \dots, \mathbf{u}_r$ gives $a_1 = \dots = a_r = 0$, proving (i).

For $\mathbf{x} \in \mathcal{V} = \mathcal{U} \oplus \ker(T)$, write

$$\mathbf{x} = \mathbf{u} + \mathbf{k}, \quad \mathbf{u} = \sum_{i=1}^r a_i \mathbf{u}_i, \quad \mathbf{k} \in \ker(T).$$

Then

$$T(\mathbf{x}) = T(\mathbf{u}) + T(\mathbf{k}) = \sum_{i=1}^r a_i T(\mathbf{u}_i).$$

Thus the images span $\text{im}(T)$; part (i) gives independence. ■

Exercise 2.10. Let $T : \mathcal{V} \rightarrow \mathcal{W}$ be linear, with $\mathcal{V}$ and $\mathcal{W}$ finite-dimensional.

- (i) If $\dim \mathcal{V} > \dim \mathcal{W}$, prove that T is not one-to-one.
- (ii) If $\dim \mathcal{V} < \dim \mathcal{W}$, prove that T is not onto.
- (iii) Deduce that for $\mathbf{A} \in \mathbb{R}^{m \times n}$, $n > m$ implies $\mathbf{Ax} = \mathbf{Ay}$ for some $\mathbf{x} \neq \mathbf{y}$, while $m > n$ implies $\mathbf{Ax} = \mathbf{b}$ is inconsistent for some $\mathbf{b} \in \mathbb{R}^m$.

Origin: H Thm. 22.1.1 and Cor. 22.1.2

Solution. Rank–nullity and $\text{im}(T) \subseteq \mathcal{W}$ give

$$\dim \ker(T) = \dim \mathcal{V} - \dim \text{im}(T) \geq \dim \mathcal{V} - \dim \mathcal{W}.$$

If $\dim \mathcal{V} > \dim \mathcal{W}$, then $\ker(T) \neq \{\mathbf{0}\}$, so T is not one-to-one. If $\dim \mathcal{V} < \dim \mathcal{W}$, then

$$\dim \text{im}(T) \leq \dim \mathcal{V} < \dim \mathcal{W},$$

so $\text{im}(T) \neq \mathcal{W}$.

For $T_{\mathbf{A}} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, if $n > m$, choose $\mathbf{h} \in \text{null}(\mathbf{A}) \setminus \{\mathbf{0}\}$ and set $\mathbf{y} = \mathbf{x} + \mathbf{h}$; then $\mathbf{A}\mathbf{y} = \mathbf{A}\mathbf{x}$. If $m > n$, choose $\mathbf{b} \in \mathbb{R}^m \setminus \text{col}(\mathbf{A})$; then $\mathbf{A}\mathbf{x} = \mathbf{b}$ is inconsistent. ■

Exercise 2.11. Let $\mathcal{B}, \mathcal{C}, \mathcal{D}$ be ordered bases of the same n -dimensional vector space. Prove

$$\mathbf{P}_{\mathcal{D} \leftarrow \mathcal{B}} = \mathbf{P}_{\mathcal{D} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}.$$

Deduce

$$\mathbf{P}_{\mathcal{B} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} = \mathbf{I}_n.$$

Origin: H Thm. 22.4.2 and §22.4b(3)

Solution. For every $\mathbf{x}$,

$$\begin{aligned} \mathbf{P}_{\mathcal{D} \leftarrow \mathcal{B}}[\mathbf{x}]_{\mathcal{B}} &= [\mathbf{x}]_{\mathcal{D}} \\ &= \mathbf{P}_{\mathcal{D} \leftarrow \mathcal{C}}[\mathbf{x}]_{\mathcal{C}} \\ &= \mathbf{P}_{\mathcal{D} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}[\mathbf{x}]_{\mathcal{B}}. \end{aligned}$$

For every $\mathbf{z} \in \mathbb{R}^n$, the vector $\mathbf{x} = \sum_{j=1}^n z_j \mathbf{v}_j$, where $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, satisfies $[\mathbf{x}]_{\mathcal{B}} = \mathbf{z}$. Hence the two matrices agree on every $\mathbf{z}$ and are equal. Setting $\mathcal{D} = \mathcal{B}$ gives

$$\mathbf{P}_{\mathcal{B} \leftarrow \mathcal{C}} \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} = \mathbf{P}_{\mathcal{B} \leftarrow \mathcal{B}} = \mathbf{I}_n.$$

Exercise 2.12. Fix distinct $r, s \in \mathbb{R}$ and define $T : \mathcal{P}_2 \rightarrow \mathbb{R}^2$ by

$$T(p) = \begin{pmatrix} p(r) \\ p(s) \end{pmatrix}.$$

Prove that T is linear and onto, find a basis of $\ker(T)$, and verify rank–nullity.

Origin: Course formulation

Solution. For $a, b \in \mathbb{R}$ and $p, q \in \mathcal{P}_2$,

$$T(ap + bq) = \begin{pmatrix} ap(r) + bq(r) \\ ap(s) + bq(s) \end{pmatrix} = aT(p) + bT(q).$$

For $(u, v)' \in \mathbb{R}^2$, define

$$p_{u,v}(t) = u \frac{t-s}{r-s} + v \frac{t-r}{s-r}.$$

Then $p_{u,v} \in \mathcal{P}_2$ and

$$T(p_{u,v}) = (u, v)',$$

so $\text{im}(T) = \mathbb{R}^2$ and $\dim \text{im}(T) = 2$. Since $\dim \mathcal{P}_2 = 3$, rank–nullity gives $\dim \ker(T) = 1$. The nonzero polynomial

$$q(t) = (t-r)(t-s)$$

satisfies $T(q) = \mathbf{0}$; hence (q) is a basis of $\ker(T)$. Therefore

$$\dim \ker(T) + \dim \text{im}(T) = 1 + 2 = 3 = \dim \mathcal{P}_2.$$



Exercise 2.13. Let $\mathbf{x}_1, \mathbf{x}_2$ be linearly independent, set $\mathbf{x}_3 = \mathbf{x}_1 - \mathbf{x}_2$, and write $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$.

(i) Find a basis of $\text{null}(\mathbf{X})$.

- (ii) For $\boldsymbol{\beta} \in \mathbb{R}^3$, find every $\boldsymbol{\gamma}$ such that $\mathbf{X}\boldsymbol{\gamma} = \mathbf{X}\boldsymbol{\beta}$.
- (iii) With $\mathbf{Z} = (\mathbf{x}_1, \mathbf{x}_2)$, find the unique $\boldsymbol{\theta} \in \mathbb{R}^2$ such that $\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}\boldsymbol{\theta}$.

Origin: Course formulation from H Lemma 4.2.2 and Thm. 11.2.3

Solution. For $\mathbf{h} = (h_1, h_2, h_3)'$,

$$\mathbf{X}\mathbf{h} = (h_1 + h_3)\mathbf{x}_1 + (h_2 - h_3)\mathbf{x}_2.$$

Independence of $\mathbf{x}_1, \mathbf{x}_2$ gives

$$\mathbf{X}\mathbf{h} = \mathbf{0} \iff h_1 = -h_3, \quad h_2 = h_3.$$

Thus

$$\text{null}(\mathbf{X}) = \text{span}\{(-1, 1, 1)'\}.$$

Consequently,

$$\mathbf{X}\boldsymbol{\gamma} = \mathbf{X}\boldsymbol{\beta} \iff \boldsymbol{\gamma} = \boldsymbol{\beta} + t(-1, 1, 1)', \quad t \in \mathbb{R}.$$

Finally,

$$\mathbf{X}\boldsymbol{\beta} = (\beta_1 + \beta_3)\mathbf{x}_1 + (\beta_2 - \beta_3)\mathbf{x}_2 = \mathbf{Z} \begin{pmatrix} \beta_1 + \beta_3 \\ \beta_2 - \beta_3 \end{pmatrix}.$$

Independence of the columns of $\mathbf{Z}$ makes the displayed coefficient vector unique. ■

Exercise 2.14. A factor has levels $1, \dots, c$, every level occurs, and $\mathbf{d}_j$ is the indicator column of level j . Set

$$\mathbf{X}_C = (\mathbf{d}_1, \dots, \mathbf{d}_c), \quad \mathbf{X}_F = (\mathbf{1}, \mathbf{d}_1, \dots, \mathbf{d}_c).$$

Fix a baseline level r and let $\mathbf{X}_{T,r}$ contain $\mathbf{1}$ followed by all $\mathbf{d}_j$ with $j \neq r$.

- (i) Find bases of $\text{null}(\mathbf{X}_F)$ and $\text{col}(\mathbf{X}_F)$.
- (ii) Prove that the columns of $\mathbf{X}_C$ and $\mathbf{X}_{T,r}$ are bases of the same model space.
- (iii) For cell means $\mu_1, \dots, \mu_c$, derive the coefficients under baseline- r treatment coding and prove equality of the two predictors.

Origin: Course formulation from H Lemma 4.2.2 and Thm. 11.2.3

Solution. Every observation belongs to exactly one level, so

$$\mathbf{1} = \sum_{j=1}^c \mathbf{d}_j, \quad \mathbf{X}_F(1, -1, \dots, -1)' = \mathbf{0}.$$

The indicator columns are independent: from $\sum_j a_j \mathbf{d}_j = \mathbf{0}$, evaluation at an observation in level j gives $a_j = 0$. Hence

$$\text{rank}(\mathbf{X}_F) = c, \quad \text{col}(\mathbf{X}_F) = \text{span}\{\mathbf{d}_1, \dots, \mathbf{d}_c\}.$$

Rank-nullity gives $\dim \text{null}(\mathbf{X}_F) = (c + 1) - c = 1$; the displayed nonzero null vector therefore gives

$$\text{null}(\mathbf{X}_F) = \text{span}\{(1, -1, \dots, -1)'\}.$$

The relations

$$\mathbf{1} = \sum_{j=1}^c \mathbf{d}_j, \quad \mathbf{d}_r = \mathbf{1} - \sum_{j \neq r} \mathbf{d}_j$$

give $\text{col}(\mathbf{X}_C) = \text{col}(\mathbf{X}_{T,r})$. If

$$a_0 \mathbf{1} + \sum_{j \neq r} a_j \mathbf{d}_j = \mathbf{0},$$

evaluation at level r gives $a_0 = 0$, and evaluation at level $j \neq r$ gives $a_0 + a_j = 0$. Thus the treatment columns are independent; both lists are bases.

The cell-means coefficients are $(\mu_1, \dots, \mu_c)'$. The treatment coefficients are

$$\alpha_0 = \mu_r, \quad \alpha_j = \mu_j - \mu_r \quad (j \neq r).$$

At level r the treatment predictor is $\alpha_0 = \mu_r$; at level $j \neq r$ it is $\alpha_0 + \alpha_j = \mu_j$. Hence both predictors agree in every component. ■

Exercise 2.15. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have full column rank. Let $\mathcal{C} = (\mathbf{c}_1, \dots, \mathbf{c}_p)$ be an ordered basis of $\mathbb{R}^p$, put $\mathbf{C} = (\mathbf{c}_1, \dots, \mathbf{c}_p)$, and define $\mathbf{Z} = \mathbf{XC}$.

- (i) Prove that $\mathbf{Z}$ has full column rank and $\text{col}(\mathbf{Z}) = \text{col}(\mathbf{X})$.
- (ii) For every $\boldsymbol{\beta} \in \mathbb{R}^p$, prove that there is exactly one $\boldsymbol{\gamma}$ such that $\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}\boldsymbol{\gamma}$, and identify $\boldsymbol{\gamma}$ as a coordinate vector.

Origin: Course formulation from H Lemma 4.2.2 and §22.4b(3)

Solution. Each column $\mathbf{X}\mathbf{c}_j$ of $\mathbf{Z}$ belongs to $\text{col}(\mathbf{X})$, so $\text{col}(\mathbf{Z}) \subseteq \text{col}(\mathbf{X})$. For $\boldsymbol{\beta} \in \mathbb{R}^p$, the basis $\mathcal{C}$ gives a unique $\boldsymbol{\gamma}$ such that

$$\boldsymbol{\beta} = \sum_{j=1}^p \gamma_j \mathbf{c}_j = \mathbf{C}\boldsymbol{\gamma}.$$

Hence

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\mathbf{C}\boldsymbol{\gamma} = \mathbf{Z}\boldsymbol{\gamma},$$

which proves $\text{col}(\mathbf{X}) \subseteq \text{col}(\mathbf{Z})$. If $\mathbf{Z}\boldsymbol{\gamma} = \mathbf{0}$, then

$$\mathbf{X}\mathbf{C}\boldsymbol{\gamma} = \mathbf{0} \implies \mathbf{C}\boldsymbol{\gamma} = \mathbf{0} \implies \boldsymbol{\gamma} = \mathbf{0},$$

using full column rank of $\mathbf{X}$ and independence of $\mathbf{c}_1, \dots, \mathbf{c}_p$. Thus $\mathbf{Z}$ has full column rank. Finally,

$$\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}\boldsymbol{\gamma} \iff \mathbf{X}(\boldsymbol{\beta} - \mathbf{C}\boldsymbol{\gamma}) = \mathbf{0} \iff \boldsymbol{\beta} = \mathbf{C}\boldsymbol{\gamma} \iff \boldsymbol{\gamma} = [\boldsymbol{\beta}]_{\mathcal{C}}.$$

